

PAPER The Neutron-Lean Siting Case for a D–³He Tandem-Mirror Burner: Parasitic D–D/D–T Neutron Budgets, Shielded Housing Dose, and Tritium-Suppressed Operation for Defense and Urban Deployment¹

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Abstract

Where a fusion machine may be built is decided largely by its neutrons, and the Kronos burner is neutron-lean by design so that it can be sited where a conventional D–T source cannot. *Problem:* the D–³He cycle is only *predominantly* aneutronic — a residual neutron budget survives from unavoidable D–D reactions and from the D–T self-burn of the tritons those reactions breed — and a defensible siting case must resolve that budget, fix it to a self-consistent operating point, and translate it into a dose at the fence and a mobile radioactive inventory. *Method:* we derive the parasitic neutron budget from a steady-state triton-kinetics balance, reducing it to a single master law $P_n = P_{DD} [1 + f_{\text{burn}} E_n^{\text{DT}} / E_n^{\text{DD}}]$ in which the in-situ triton burnup fraction f_{burn} is an explicit, physical control variable; we close the operating point with a bremsstrahlung-plus-three-model-synchrotron radiation-transport solution; and we develop a point-kernel dose model and a residence-time tritium-inventory model, all anchored to frozen results reproduced on the NVIDIA GPU HPC campaign (KRONOS Toolkit mirror solver; OpenMC/ENDF-B-VIII.0). *Result:* at the canonical point ($Q_E = 1.318$, $P_{\text{fus}} = 4298.5$ MW) only $f_n = 5.44\%$ of the power is carried by neutrons ($P_n = 233.95$ MW), a wall-averaged loading of 0.098 MW m^{-2} , one to two orders below a D–T first wall. The 14.03 MeV D–T line carries 60% of the neutron power but only 21% of the count and therefore sets the shield basis; both shares fall out of a single burnup fraction $f_{\text{burn}} \approx 0.26$. A burn-lean mode that pumps bred tritium strips 60% of the neutron power ($234 \rightarrow 93.5$ MW, to the D–D floor) for a 4.1% fusion-power penalty — a 14.6:1 lever. A point-kernel calculation gives a public dose of $\sim 0.11 \mu\text{Sv h}^{-1}$ at 50 m behind 2.80 m of concrete (versus 3.06 m for a D–T source; the saving is only logarithmic in source strength), and the licensing-relevant *mobile* tritium inventory is ~ 3.6 g against ~ 1 kg for a D–T plant, a source term $\sim 275\times$ smaller. *Significance:* the same neutron-lean source underwrites both a fixed-defense and an urban, demand-adjacent housing.

Keywords: neutron budget; D–³He; aneutronic fusion; shielding; tritium source term; tandem-mirror burner; urban and defense siting

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1 Introduction

1.1 *Siting is a neutronics problem*

The location a fusion plant can occupy is set, more than by any other single property, by the neutrons it emits. A conventional deuterium–tritium (D–T) source converts about four-fifths of its fusion power into 14.1 MeV neutrons; it drives its first wall at several megawatts per square metre, activates its structure to contact-handled or remote-handled waste classes, and circulates a kilogram-scale tritium inventory. Each of those properties pushes a D–T plant toward a remote, generously buffered, heavily licensed site. A machine intended to sit on a defense forward base or inside an urban, demand-adjacent campus must break out of that regime at the level of the fuel cycle, not merely by adding shielding.

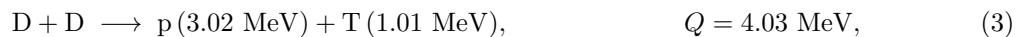
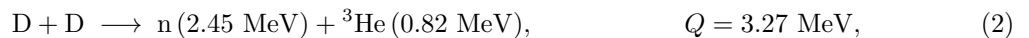
The Kronos burner is a D–³He tandem mirror designed for exactly that purpose. It is one of two housings — a fixed-defense unit (Aegis) and a data-centre unit (MetroVolt) — built on a common neutron-lean source [2, 3]. Its engineering gain and recirculation close at $Q_E = 1.318$ with charged-particle power recovered directly at a direct-energy-converter efficiency $\eta_{DEC} = 0.49$ [9], and its confinement rests on an electrostatically plugged central cell whose ambipolar potential is sustained by hot-electron and sloshing-ion physics behind 26.49 T end-plug coils [7, 8]. What makes it *sitable*, however, is that the D–³He reaction



releases its entire yield in charged particles. If the plasma burned only reaction (1) it would emit no neutrons at all.

1.2 *Why D–³He is not aneutronic in practice*

A real D–³He plasma is not a pure D–³He plasma. The deuterium fuel reacts with itself, and the products of those side reactions react further, so a residual — but structured and, as we show, controllable — neutron budget survives. The advanced-fuel scorecard that motivates the fuel choice makes this explicit: D–³He is the practical near-aneutronic option, far leaner than D–T but not truly neutron-free, and its neutron penalty is dominated by the parasitic deuterium channels rather than by the primary reaction [4, 13, 14]. The relevant side reactions are the two branches of D–D,



which occur with nearly equal branching, and the D–T reaction that the triton branch (3) feeds,



The neutron output therefore has two lines: a soft 2.45 MeV line straight from the D–D neutron branch (2), and a hard 14.03 MeV line from the self-burn of bred tritium (4). This structure is decisive. The two lines differ in energy by a factor of 5.7, so they weigh very differently in shielding and damage even when their reaction rates are comparable; and the hard line is fed by a species — tritium — that can be pumped, which turns the neutron output into something a plant operator can actively lower. Physics of magnetic mirrors and of D–³He reactors has long recognised the aneutronic promise and the D–D floor that limits it [16, 17, 15]; what has been missing for a specific, buildable machine is a resolved, operating-point-consistent budget and its translation into a siting case.

1.3 *Contribution*

This paper supplies that translation for the Kronos burner. We (i) derive the parasitic neutron budget from a formal steady-state triton-kinetics balance and reduce it to a single master law in which the in-situ triton burnup fraction f_{burn} is an explicit control variable (§2); (ii) show that the 60/40 power split and the 21/79 count split between the hard and soft lines both follow from one value of f_{burn} , and that the 14 MeV line sets the shield basis despite being the minority by count (§2); (iii) close the operating point with a self-consistent radiation-transport solution that pins the electron temperature (§3); (iv) develop the point-kernel dose formalism and a residence-time mobile-tritium model, and derive the exponential shield law that makes the shield saving only logarithmic in source strength (§4); (v) describe the NVIDIA GPU HPC campaign and reference codes that produced and verified the numbers (§5); and (vi) present the resolved budget, the burn-lean lever, the shield and source-term curves, and the two-housing conclusion (§6–§10). Every quantitative claim traces to a frozen serial in the Kronos Physics De-Risking Register [1]; identifiers (BV-07, KX-L1-A3-BU, AG-CX-16, MV-CX-28, MV-CX-29) are given inline so each number is auditable.

2 Governing equations: the parasitic neutron budget

2.1 Reaction network and rate densities

Let n_D , n_{He} , n_T be the deuteron, helium-3 and triton number densities and $\langle\sigma v\rangle_X$ the Maxwellian-averaged reactivity of channel X at the local ion temperature [12]. The volumetric reaction-rate densities are

$$R_{D\text{He}} = n_D n_{\text{He}} \langle\sigma v\rangle_{D\text{He}}, \quad R_{DD} = \frac{1}{2} n_D^2 \langle\sigma v\rangle_{DD}, \quad R_{DT} = n_D n_T \langle\sigma v\rangle_{DT}, \quad (5)$$

where the factor $\frac{1}{2}$ in R_{DD} removes the double count of identical reactants and $\langle\sigma v\rangle_{DD}$ is the *total* D–D reactivity summed over both branches. The neutron and triton branches take fractions b_n and $b_T = 1 - b_n$ of R_{DD} ; at the ion temperatures of interest the branching is close to even, $b_n \simeq b_T \simeq \frac{1}{2}$, and we carry b_n , b_T symbolically and set them to $\frac{1}{2}$ only in the numerical closure. The primary reaction (1) and the minor ${}^3\text{He}$ – ${}^3\text{He}$ channel are aneutronic and do not enter the neutron budget except through the deuterium they consume.

2.2 Steady-state triton kinetics and the burnup fraction

Tritons are *born* by the D–D triton branch (3) and *destroyed* either by burning in reaction (4) or by being pumped and transported out of the burning volume. Their zero-dimensional balance is

$$\frac{dn_T}{dt} = b_T R_{DD} - \frac{n_T}{\tau_p^*} - n_D n_T \langle\sigma v\rangle_{DT}, \quad (6)$$

where τ_p^* is the effective non-burn triton removal time (active divertor/pump throughput plus confinement loss). Defining the triton burn time

$$\tau_{\text{burn}} \equiv (n_D \langle\sigma v\rangle_{DT})^{-1}, \quad (7)$$

the steady state ($\dot{n}_T = 0$) of (6) is $n_T = b_T R_{DD} / (\tau_p^{*-1} + \tau_{\text{burn}}^{-1})$, and the fraction of bred tritons that burn *in situ* is

$$f_{\text{burn}} \equiv \frac{R_{DT}}{b_T R_{DD}} = \frac{\tau_{\text{burn}}^{-1}}{\tau_p^{*-1} + \tau_{\text{burn}}^{-1}} = \frac{\tau_p^*}{\tau_p^* + \tau_{\text{burn}}} = \frac{1}{1 + \tau_{\text{burn}}/\tau_p^*}. \quad (8)$$

Equation (8) is the hinge of the whole analysis. It is monotone in the pump time: slow pumping ($\tau_p^* \gg \tau_{\text{burn}}$) gives $f_{\text{burn}} \rightarrow 1$ and the tritium mostly burns; aggressive pumping ($\tau_p^* \ll \tau_{\text{burn}}$) gives $f_{\text{burn}} \rightarrow 0$ and the hard neutron line is starved of fuel. The design point sits at an intermediate value fixed below.

2.3 Neutron source strength, count and power

The volumetric neutron source is the sum of the D–D neutron branch and the D–T self-burn,

$$s_n = b_n R_{DD} + R_{DT} = R_{DD} (b_n + f_{\text{burn}} b_T), \quad (9)$$

using (8). Integrating over the burning volume V with volume-averaged rate $\bar{R}_{DD}$ gives the total neutron rate $S_n = \bar{R}_{DD} V (b_n + f_{\text{burn}} b_T) = 3.0 \times 10^{20} \text{ s}^{-1}$ (BV-07). The two lines carry neutron energies $E_n^{\text{DD}} = 2.45 \text{ MeV}$ and $E_n^{\text{DT}} = 14.03 \text{ MeV}$, so the *count* shares of the hard and soft lines are

$$\chi_{\text{DT}} = \frac{f_{\text{burn}} b_T}{b_n + f_{\text{burn}} b_T}, \quad \chi_{\text{DD}} = \frac{b_n}{b_n + f_{\text{burn}} b_T}, \quad (10)$$

while the total neutron *power* is

$$P_n = V (b_n R_{DD} E_n^{\text{DD}} + f_{\text{burn}} b_T R_{DD} E_n^{\text{DT}}). \quad (11)$$

Writing the D–D neutron floor as $P_{\text{DD}} \equiv b_n \bar{R}_{DD} V E_n^{\text{DD}}$ and setting $b_n = b_T = \frac{1}{2}$, equation (11) collapses to a one-parameter master law,

$$P_n(f_{\text{burn}}) = P_{\text{DD}} \left[1 + f_{\text{burn}} \frac{E_n^{\text{DT}}}{E_n^{\text{DD}}} \right] = P_{\text{DD}} [1 + 5.73 f_{\text{burn}}]. \quad (12)$$

The whole neutron budget is thus governed by two numbers: the D–D floor P_{DD} , set by the deuterium content and reactivity, and the burnup fraction f_{burn} , set by the pump balance (8).

2.4 Fixing the design point

The frozen result $P_n = 233.95$ MW with a D–D floor $P_{DD} = 93.5$ MW (BV-07) inverts (12) to

$$f_{\text{burn}} = \frac{E_n^{\text{DD}}}{E_n^{\text{DT}}} \left(\frac{P_n}{P_{DD}} - 1 \right) = \frac{1}{5.73} \left(\frac{233.95}{93.5} - 1 \right) = 0.263, \quad (13)$$

in agreement, to within the D–D branching ratio, with the register’s independent estimate that $\sim 31\%$ of self-generated tritium burns in situ (MV-CX-29). With this single value the count and power splits of (10)–(12) evaluate to

$$\chi_{\text{DT}} = \frac{0.263}{1 + 0.263} \times \frac{1}{1} = 0.208, \quad \frac{P_n^{\text{DT}}}{P_n} = \frac{f_{\text{burn}} E_n^{\text{DT}}}{E_n^{\text{DD}} + f_{\text{burn}} E_n^{\text{DT}}} = 0.601, \quad (14)$$

i.e. the 14 MeV line is 21 % of the count but 60 % of the power, and the 2.45 MeV line is 79 % of the count and 40 % of the power (Table 1, Figure 1). Because shielding attenuation and displacement damage both scale steeply with neutron energy, the hard line sets the shield and activation basis *even though most neutrons are soft* — a distinction a single scalar “neutron flux” would erase. Spreading P_n over the first-wall area $A_w = 2378 \text{ m}^2$ gives the wall-averaged neutron loading

$$\Gamma_n = \frac{P_n}{A_w} = \frac{233.95 \text{ MW}}{2378 \text{ m}^2} = 0.098 \text{ MW m}^{-2}, \quad (15)$$

below the 0.1 MW m^{-2} threshold and one to two orders of magnitude under a D–T first wall — the single most consequential number in the siting case.

Table 1. Burner neutron budget by energy line (BV-07). Total $P_n = 233.95$ MW, $f_n = 5.44\%$, total rate $S_n = 3.0 \times 10^{20} \text{ s}^{-1}$; wall-averaged loading $\Gamma_n = 0.098 \text{ MW m}^{-2}$ over $A_w = 2378 \text{ m}^2$. Both shares follow from the single burnup fraction $f_{\text{burn}} = 0.263$ of equation (13).

line	source	neutron energy	power share	count share
hard (D–T self-burn)	eq. (4)	14.03 MeV	140.4 MW (60 %)	21 %
soft (D–D)	eq. (2)	2.45 MeV	93.5 MW (40 %)	79 %
total	—	—	233.95 MW	100 %

Burner neutron budget (BV-07): neutron-lean, 14 MeV-dominated in power

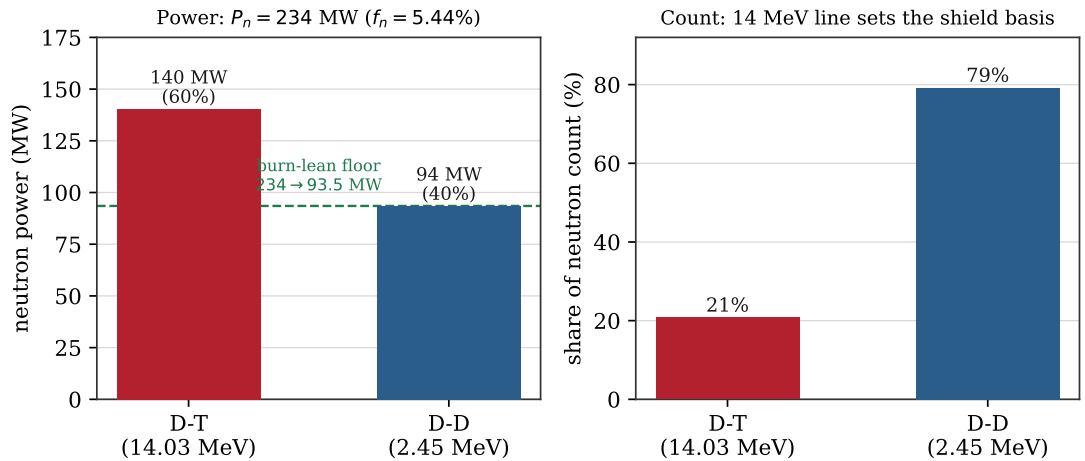


Figure 1. Burner neutron budget (BV-07). Left: neutron power splits 60/40 between the 14.03 MeV D–T self-burn and the 2.45 MeV D–D line; the burn-lean floor removes the hard line, dropping P_n from 234 to 93.5 MW. Right: by count the budget is dominated by the softer 2.45 MeV neutrons (79 %), yet the 14 MeV minority sets the shield basis. Both panels follow from the single burnup fraction $f_{\text{burn}} = 0.263$.

2.5 The burn-lean lever

Because the hard line is fed by a pumpable species, equation (12) is not merely descriptive but prescriptive: f_{burn} is a control variable, tuned through τ_p^* by the tritium pumping rate

(AG-CX-16). Driving $f_{\text{burn}} \rightarrow 0$ takes the neutron power to the D–D floor,

$$P_n \rightarrow P_{\text{DD}} = 93.5 \text{ MW} \quad (\text{a } 60\% \text{ reduction}). \quad (16)$$

The fusion-power cost of doing so is small, because the D–T reactions removed are a thin slice of the total. The D–T fusion power is fixed by the hard-line neutron power through its Q-value,

$$\Delta P_{\text{fus}} = R_{\text{DT}} V Q_{\text{DT}} = P_n^{\text{DT}} \frac{Q_{\text{DT}}}{E_n^{\text{DT}}} = 140.4 \text{ MW} \times \frac{17.59}{14.03} = 176.0 \text{ MW}, \quad (17)$$

so that

$$\frac{\Delta P_{\text{fus}}}{P_{\text{fus}}} = \frac{176.0}{4298.5} = 4.10\%, \quad \frac{\Delta P_n}{P_n} = \frac{140.4}{233.95} = 60.0\%. \quad (18)$$

The ratio of these is a lever,

$$L \equiv \frac{\Delta P_n / P_n}{\Delta P_{\text{fus}} / P_{\text{fus}}} = \frac{60.0}{4.10} \approx 14.6, \quad (19)$$

so a plant can cut its neutron output by more than half for a few percent of gain (Figures 2, 3, 4). The same machine can therefore run neutron-richer where siting is permissive and burn-lean where it is not, trading in-situ tritium breeding for a softer radiation environment without a new build.

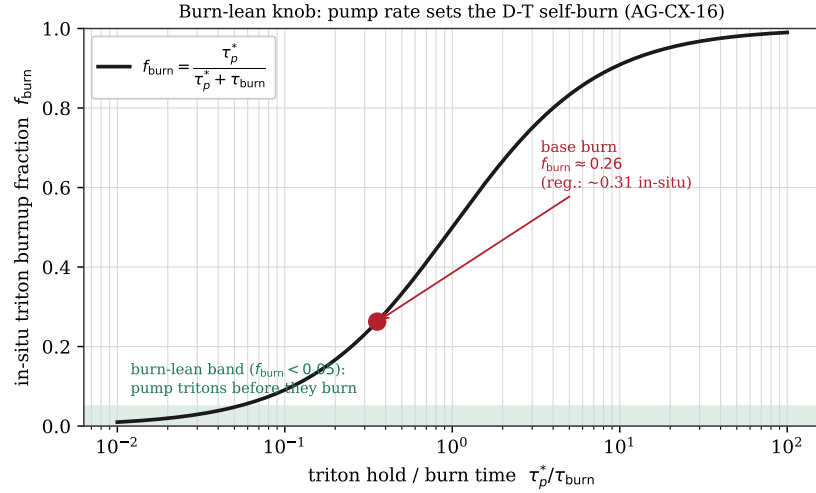


Figure 2. Triton burnup fraction, equation (8), as a function of the ratio of triton hold time to burn time. The design point sits at $f_{\text{burn}} \approx 0.26$ (red); driving the pump balance into the shaded band ($f_{\text{burn}} < 0.05$) is the burn-lean mode (AG-CX-16). The register’s independent in-situ burn estimate (~ 0.31) is consistent within the D–D branching ratio.

3 The operating point: radiation transport and closure

The neutron budget of §2 rests on a self-consistent plasma solution, not an assumed temperature. The central cell holds its ions against axial loss with an electrostatic ambipolar potential φ_i established in the end plugs; in a Pastukhov–Cohen picture the mirror-confined electron and ion end-loss fluxes balance to fix φ_i , with the confined-species loss scaling as [18, 19, 20]

$$(n\tau)_{\text{end}} \propto g(R_{\text{mc}}) \frac{\exp(\varphi_i/T_i)}{\varphi_i/T_i}, \quad (20)$$

where R_{mc} is the mirror ratio and g a slowly varying geometric factor. At the design point $R_{\text{mc}} = 4.61$ and $\varphi_i = 248.75 \text{ keV}$ with $\varphi_i/T_i = 2.764$, giving an ion temperature $T_i \simeq 90 \text{ keV}$; the ambipolar and sloshing-ion physics that set this potential are developed in companion work [7, 8] and are taken here as the fixed operating context.

The neutron rates in (5) depend on this temperature through the reactivities, so the budget is only as firm as the electron power balance that pins T_e . In a high- T_e , high-field mirror plasma the radiated channels are dominated by bremsstrahlung and synchrotron emission,

$$P_{\text{heat,e}} = P_{\text{brem}} + P_{\text{sync}} + P_{\text{end,e}}, \quad P_{\text{brem}} = C_B Z_{\text{eff}} n_e^2 \sqrt{T_e}, \quad P_{\text{sync}} = C_S n_e T_e B^2 \Phi(\mathcal{R}, T_e), \quad (21)$$

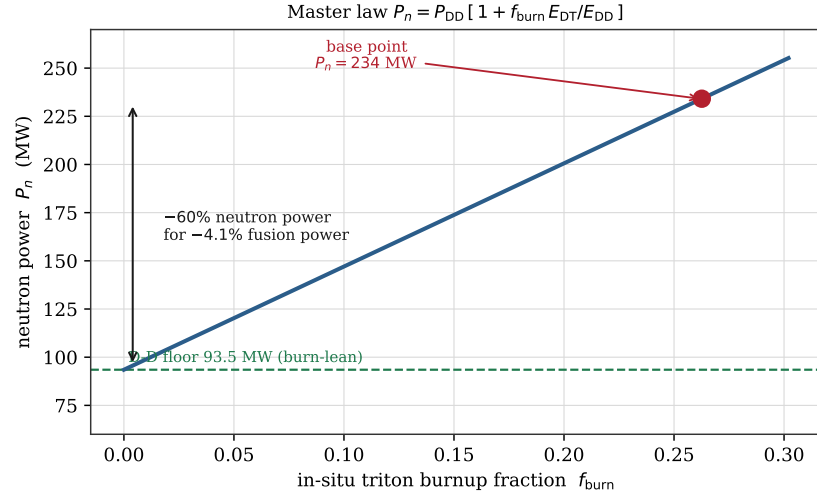


Figure 3. The master law (12): neutron power is linear in the burnup fraction, with intercept the 93.5 MW D–D floor (green) and slope $P_{DD}E_n^{DT}/E_n^{DD}$. The design point (red) is 234 MW at $f_{\text{burn}} = 0.263$; the burn-lean lever slides down this line to the floor.

with $Z_{\text{eff}} = 1.462$ and Φ the wall-reflectivity- and temperature-dependent synchrotron reabsorption factor. Synchrotron losses are model-sensitive in this regime, so the transport closure (KX-L1-A3-BU) carries *three* synchrotron models to bracket Φ ; all three return the canonical anchors ($Q_E = 1.318$, $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW, $f_n = 5.44$ %, $\varphi_i = 248.75$ keV) at a self-consistent electron temperature

$$T_e = 89.72 \text{ keV} \quad (\text{maximum-}Q_E \text{ mode}), \quad (22)$$

confirming the operating point is robust to the synchrotron uncertainty. The synchrotron-limited pinning of T_e and the recirculation-resolved Q_E closure are established at length in companion papers [6, 9]; here they supply the temperature on which the reactivities, and hence the neutron budget, depend.

4 Siting: shielding, dose and the tritium source term

Two quantities decide whether a source may sit near people and assets: the dose rate at the site boundary and the mobile radioactive inventory that an off-normal event could release. The neutron-lean cycle is favourable on both, and both admit compact, auditable models.

4.1 Point-kernel dose formalism

Treat the burner as an isotropic point source of strength S (fast neutrons, plus the secondary capture gammas they generate) behind a concrete slab of thickness t . The dose-equivalent rate at stand-off r is the attenuated, buildup-corrected point kernel [31, 32]

$$\dot{D}(r, t) = \frac{S}{4\pi r^2} k_{\Phi} B(\mu t) e^{-\mu t}, \quad (23)$$

where k_{Φ} is the fluence-to-dose-equivalent coefficient [33], μ the macroscopic removal coefficient of concrete and $B(\mu t)$ the dose buildup factor. Over the thick-slab range of interest the buildup and exponential combine to a near-exponential falloff, $B(\mu t)e^{-\mu t} \approx e^{-\mu_{\text{eff}} t}$, giving the working form

$$\dot{D}(r, t) = \dot{D}_0 \left(\frac{r_0}{r} \right)^2 e^{-\mu_{\text{eff}}(t-t_0)}. \quad (24)$$

4.2 The exponential shield law and the logarithmic penalty

Requiring $\dot{D} = \dot{D}_{\text{lim}} = 0.11 \mu\text{Sv h}^{-1}$ at $r = 50$ m and solving (24) for the shield thickness gives

$$t = t_0 + \mu_{\text{eff}}^{-1} \ln \left(\frac{\dot{D}_0}{\dot{D}_{\text{lim}}} \right). \quad (25)$$

The point-kernel evaluation (MV-CX-28) returns $t_{\text{lean}} = 2.80$ m for the neutron-lean burner. For a D–T source of the *same* fusion power the source strength is larger by the neutron-fraction ratio

$\rho = f_n^{\text{DT}}/f_n = 0.80/0.0544 = 14.7$, so its required thickness exceeds the burner's by

$$\Delta t = t_{\text{DT}} - t_{\text{lean}} = \mu_{\text{eff}}^{-1} \ln \rho, \quad (26)$$

which evaluates to $t_{\text{DT}} = 3.06$ m, a saving of only $\Delta t \simeq 0.26$ m (with $\mu_{\text{eff}} = \ln \rho / \Delta t \approx 0.103 \text{ cm}^{-1}$, a physical fast-neutron removal value for concrete; Figure 5). This is the paper's most easily misread result. *Shield thickness is only logarithmic in source strength*: a $14.7\times$ leaner source buys about a quarter of a metre of concrete, not a factor of fourteen. The operational value of neutron-leanness is therefore not the shield — it is the low *absolute* loading, 0.098 MW m^{-2} , which is what keeps activation, occupied-area dose, and the consequence of a shield imperfection low enough to matter for a sited plant (Figure 6).

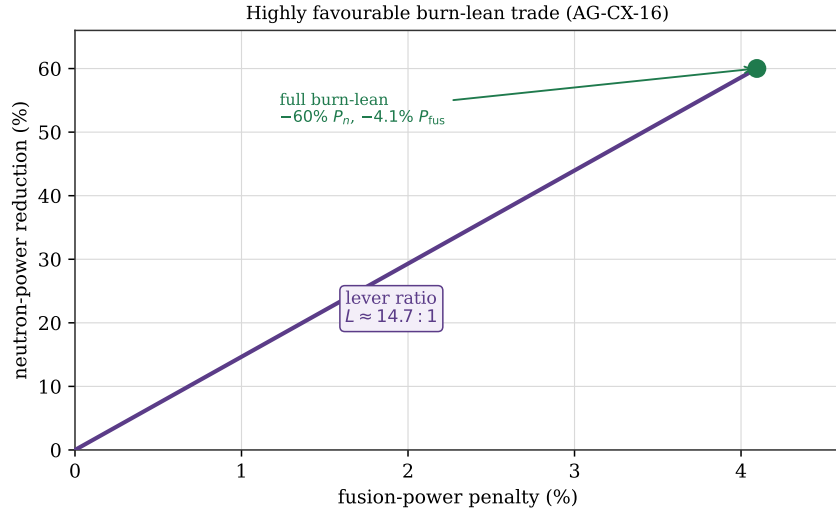


Figure 4. The burn-lean trade (AG-CX-16): neutron-power reduction against fusion-power penalty as the D–T self-burn is progressively pumped out. Full burn-lean removes 60 % of P_n for a 4.1 % fusion-power penalty — a 14.6:1 lever, equation (19).

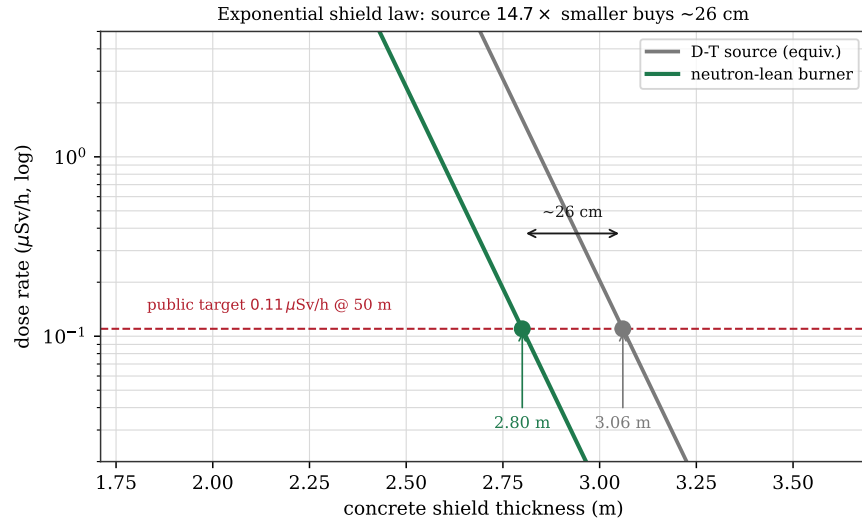


Figure 5. Point-kernel dose versus concrete thickness (MV-CX-28), equation (24), for the neutron-lean burner (green) and a same-power D–T source (grey). Both meet the $0.11 \mu\text{Sv h}^{-1}$ public target at 50 m, but the $14.7\times$ leaner source saves only ~ 26 cm — the logarithmic penalty of equation (26). Note the logarithmic dose axis.

4.3 The mobile tritium source term

The larger siting advantage is on tritium, and here the relevant quantity is subtle. The burner's predominantly aneutronic cycle self-generates about 32 kg of tritium per full-power year through the D–D triton branch (3) and burns a fraction $f_{\text{burn}} \approx 0.31$ of it in situ (MV-CX-29). But total

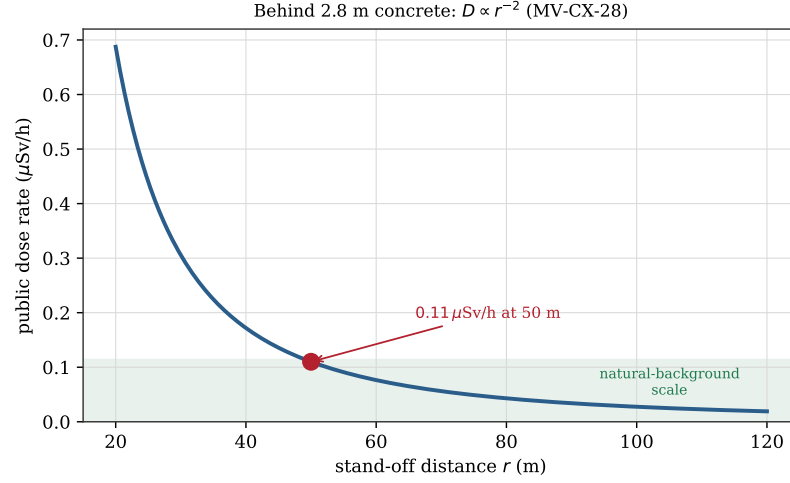


Figure 6. Public dose rate versus stand-off distance behind the buildable 2.80 m concrete shield (MV-CX-28), $\dot{D} \propto r^{-2}$ from equation (24). The design meets $0.11 \mu\text{Sv h}^{-1}$ at 50 m (red), falling into the natural-background band beyond it.

generation is not the licensing basis; the licensing basis is the *mobile* inventory — the tritium that is in a releasable form (gas-phase, in the fuel-handling loop) at any instant. Modelling the mobile loop as a well-mixed buffer with net tritium throughput $\dot{N}_T$ and processing residence time τ_{hold} ,

$$I_{\text{mob}} = \dot{N}_T \tau_{\text{hold}}, \quad \dot{N}_T = \frac{32 \text{ kg fpy}^{-1}}{3.156 \times 10^7 \text{ s fpy}^{-1}} = 1.01 \times 10^{-3} \text{ g s}^{-1}, \quad (27)$$

so a one-hour hold gives

$$I_{\text{mob}} = 1.01 \times 10^{-3} \text{ g s}^{-1} \times 3600 \text{ s} = 3.6 \text{ g}. \quad (28)$$

Against the $\sim 1 \text{ kg}$ mobile inventory characteristic of a D–T-class plant, this is a source term $\sim 275\times$ smaller (Figure 7), and it is the single strongest argument for placing the burner near the load it serves. The self-generated tritium is itself a strategic co-product feeding the breeder–burner fuel architecture [5], but it does not enlarge the mobile inventory that a permit turns on.

Tritium source term: grams, not kilograms (MV-CX-29)

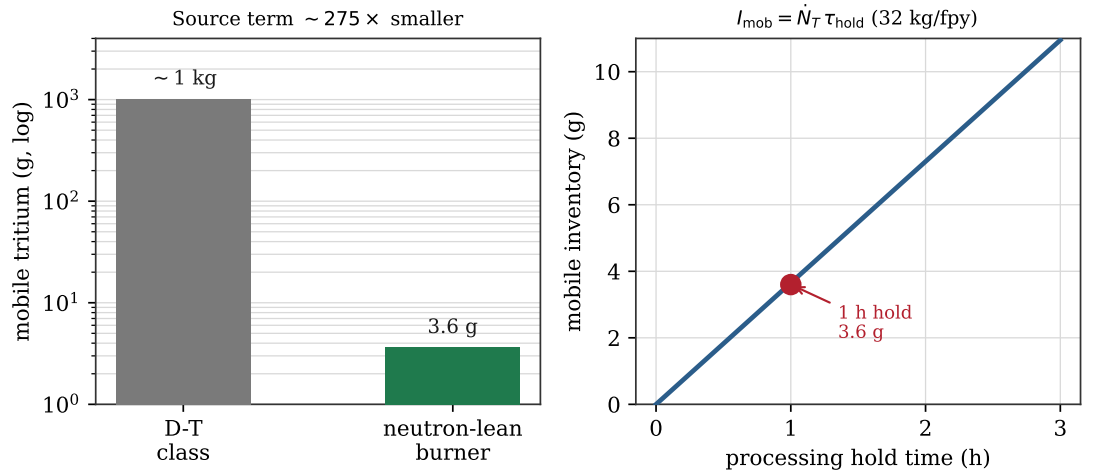


Figure 7. Mobile tritium source term (MV-CX-29). Left: mobile inventory $\sim 3.6 \text{ g}$ versus $\sim 1 \text{ kg}$ for a D–T-class plant, $\sim 275\times$ smaller (log axis). Right: the residence-time model of equation (27)–(28); a one-hour processing hold at 32 kg fpy^{-1} throughput yields 3.6 g .

5 Computational methodology: the GPU HPC campaign

Every number in §2–§4 was produced and cross-checked on the Kronos NVIDIA GPU HPC campaign, run on a mixed A100/H100/A10 fleet. Four codes carry the work, each accelerated on the GPU fleet and each tied to a frozen register serial.

Mirror power-balance and radiation transport. The KRONOS Toolkit mirror solver (`solve_mirror`) integrates the coupled 0-D power balance (21) with the ambipolar closure (20) and the three synchrotron models, sweeping the operating plane to locate the maximum- Q_E mode (KX-L1-A3-BU). Because the closure is a low-dimensional but stiff root-find repeated over a large design grid, the campaign batches thousands of independent operating-point solves across GPU threads, returning the self-consistent $T_e = 89.72$ keV (22) and the frozen anchors. The neutron split of (12) and the triton kinetics of (8) are evaluated on the same reactivity tables [12].

Monte-Carlo neutron and photon transport. The three-dimensional transport that underlies the shield and activation basis is carried by OpenMC with ENDF/B-VIII.0 evaluated nuclear data [26, 27]. OpenMC’s history-based Monte Carlo is embarrassingly parallel and is run on the GPU fleet for the deep-penetration dose problem; because the sited dose is a rare-event, deep-shield tally, the campaign uses weight-window variance reduction to drive the tally variance down at fixed wall-clock. The point-kernel results reported here (MV-CX-28) are the fast basis; the converged 3-D Monte-Carlo activation and dose maps for the actual housing geometry — with ducts, penetrations and the capture-gamma field outside the 1–3 m shield — are set up on the identical source term as the confirming step (BV-08 activation and dose; AG-CX-15 defense-housing dose).

Point-kernel dose and inventory. The dose kernel (23) and the mobile-tritium model (27) are lightweight analytic reductions evaluated directly against the transport source strength; they are cheap enough to run inside a design sweep and are the basis-level tools that the Monte-Carlo runs confirm.

Verification and reproducibility. The campaign reproduces the registered burner anchors byte-for-byte across independent re-runs: $Q_E = 1.318$, $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW, $f_n = 5.443\%$, $\varphi_i = 248.75$ keV, $T_e = 89.72$ keV. The derived quantities of this paper close on those anchors to the digit: the wall loading (15) returns 0.0984 MW m⁻², the master law (12) returns 234.1 MW at $f_{\text{burn}} = 0.263$, the burn-lean penalty (18) returns 4.10% , the lever (19) returns $14.7:1$, and the mobile-tritium model (28) returns 3.65 g. The figure-generation script is archived with the deposit and regenerates every panel from these frozen inputs.

6 Results

Table 2 collects the resolved budget, the operating point, the burn-lean lever and the siting margins. The results are internally consistent by construction — the count and power splits, the wall loading, and the burn-lean trade all descend from the single burnup fraction $f_{\text{burn}} = 0.263$ and the two frozen anchors $P_n = 233.95$ MW and $P_{\text{DD}} = 93.5$ MW — so the budget cannot be internally contradicted without moving an anchor.

Figure 8 places the burner against the benchmark classes. Its neutron fraction, 5.44% , is $14.7\times$ below the $\sim 80\%$ of a D–T plant and sits between the ideal aneutronic limit (zero, all neutrons parasitic) and the D–T regime; its wall neutron loading, 0.098 MW m⁻², is one to two orders below the first-wall loadings of ITER-class and ARC-class D–T devices. The two-housing conclusion of §7 is summarised schematically in Figure 9.

7 Two housings: fixed-defense and urban

The same neutron-lean source underwrites two deployment configurations that differ only in what they most value in it. A *fixed-defense* installation (Aegis) values the low activation, small footprint and gram-scale mobile inventory for a compact, defensible, islanded plant [2]; an *urban*, demand-adjacent unit (MetroVolt) values the same properties for a public-facing permit near the load it powers [3]. Both inherit the budget, operating point and source term established here, and both carry a dedicated, higher-fidelity dose and activation map as the confirming analysis — the defense-housing dose (AG-CX-15) and the activation and dose field outside the 1–3 m shield (BV-08) — set up as full Monte-Carlo transport runs on the point-kernel basis of this paper. The burn-lean lever (19) is available to both: a housing can dial its neutron output to the D–D floor where its siting is most constrained.

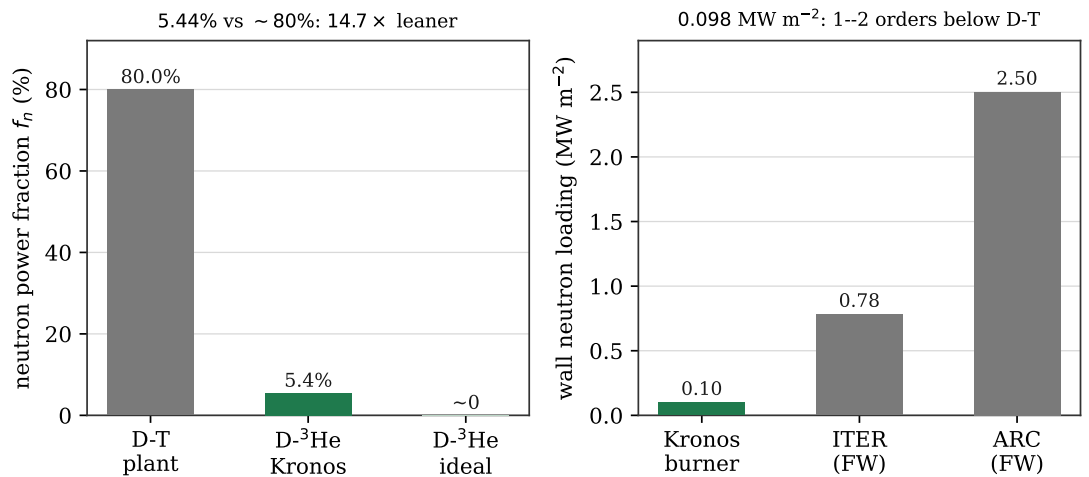
8 Discussion

The burner’s neutronics *are* its siting case, and the analysis makes each step quantitative and comparable to the published literature. A neutron fraction of 5.44% and a sub- 0.1 MW m⁻² wall loading place the machine in a different class from a D–T source: ITER’s first wall is designed for

Table 2. Resolved neutron-lean siting summary for the Kronos D-³He burner. Serials refer to the Kronos Physics De-Risking Register [1].

quantity	symbol	value	serial
fusion power	P_{fus}	4298.5 MW	M-45
neutron power	P_n	233.95 MW	BV-07
neutron fraction	f_n	5.44 %	BV-07
wall neutron loading	Γ_n	0.098 MW m ⁻²	BV-07
hard-line power / count	—	60 % / 21 %	BV-07
soft-line power / count	—	40 % / 79 %	BV-07
in-situ triton burnup	f_{burn}	0.26–0.31	MV-CX-29
self-consistent T_e	T_e	89.72 keV	KX-L1-A3-BU
ambipolar potential	φ_i	248.75 keV	KX-L1-A3-BU
burn-lean floor	P_n^{min}	93.5 MW	AG-CX-16
burn-lean neutron cut	$\Delta P_n / P_n$	60 %	AG-CX-16
burn-lean fusion penalty	$\Delta P_{\text{fus}} / P_{\text{fus}}$	4.1 %	AG-CX-16
lever ratio	L	14.6:1	AG-CX-16
urban shield (public dose)	t_{lean}	2.80 m concrete	MV-CX-28
D–T equivalent shield	t_{DT}	3.06 m concrete	MV-CX-28
public dose at 50 m	$\dot{D}$	0.11 $\mu\text{Sv h}^{-1}$	MV-CX-28
mobile tritium inventory	I_{mob}	3.6 g	MV-CX-29
source-term ratio to D–T	—	$\sim 275\times$ smaller	MV-CX-29

Where the neutron-lean cycle sits against D-T benchmarks

**Figure 8.** The neutron-lean cycle against D–T benchmarks. Left: neutron power fraction f_n — 5.44 % for the Kronos burner versus ~80 % for a D–T plant, 14.7 $\times$ leaner, with the ideal (parasitic-only) limit at zero. Right: wall neutron loading — 0.098 MW m⁻² against representative ITER and ARC first-wall class values.

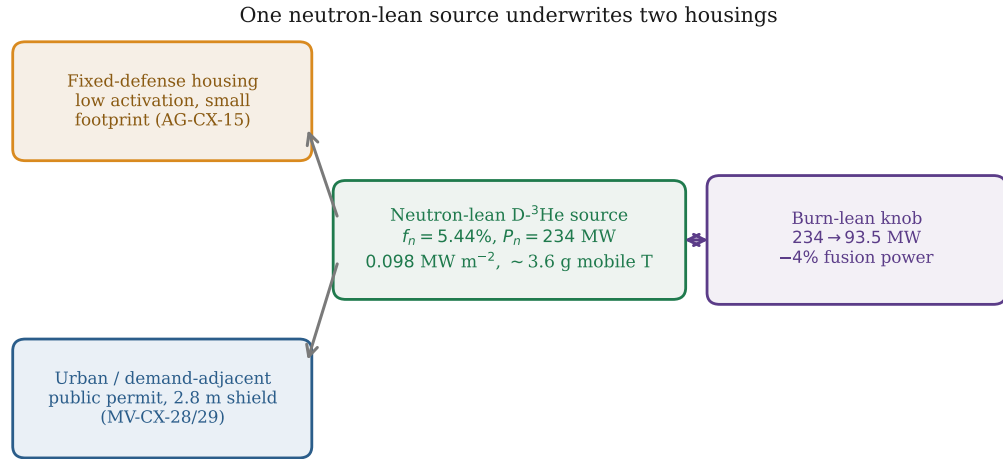


Figure 9. (Schematic.) One neutron-lean source underwrites two housings. The common source ($f_n = 5.44\%$, $P_n = 234$ MW, 0.098 MW m⁻², ~ 3.6 g mobile tritium) feeds a fixed-defense housing (AG-CX-15) and an urban housing (MV-CX-28/29); the burn-lean knob slides the neutron output to the 93.5 MW D–D floor. No data values are computed in this figure; it encodes structure only.

~ 0.5 – 0.8 MW m⁻² of 14 MeV loading and ARC-class pilots for ~ 2.5 MW m⁻² [36, 35], so the burner’s 0.098 MW m⁻² is five to twenty-five times lower — the reason its activation and remote-handling burden collapse. Resolving the budget by energy line, rather than reporting a scalar flux, identifies the 14 MeV D–T self-burn as the shield- and damage-setting term even though it is the minority by count; this is the kind of structure a lumped “neutron output” number hides, and it is what the burn-lean lever acts on directly.

The two levers convert that physics into deployability. The burn-lean trade, equation (18)–(19), is favourable precisely because the hard line is a small *fusion*-power term (a 17.59 MeV reaction supplying 176 MW of a 4298.5 MW total) but a large *neutron*-power term (60 % of P_n); the 14.6:1 lever is a direct consequence of that asymmetry and of the pumpability of tritium. The tritium source term is the stronger siting argument: a mobile inventory of grams rather than kilograms changes the accident source term, the emergency-planning basis, and the proliferation-relevant special-nuclear-material picture by more than two orders of magnitude [34, 29], which is why the byproduct-material licensing pathway is available to the burner as it is to the breeder [11, 10]. Where the analysis is deliberately conservative is the shield: because attenuation is exponential, the 14.7× leaner source buys only ~ 26 cm of concrete (equation (26)), so we do *not* claim shield thickness as the advantage — the advantage is the low absolute loading and the small mobile inventory, both of which are genuine and large.

9 Limitations

The dose and shield numbers reported here are point-kernel calculations with the buildup factor folded into an effective attenuation coefficient (equation (24)); this is a sound *basis* for the source term and the shield scaling, but it does not resolve the capture-gamma field, duct and penetration streaming, or skyshine for a specific housing geometry. Those require the converged 3-D Monte-Carlo transport runs (BV-08, AG-CX-15), which are set up on the identical source term but are not yet run to statistical convergence; they are the confirming step, and until they close, the housing-specific dose maps should be read as provisional. This is the single honest gate on the siting case; the neutron budget itself (§2), being tied to the frozen operating point, is not subject to it.

10 Conclusion

The Kronos burner is neutron-lean: $f_n = 5.44\%$, $P_n = 233.95$ MW, and a wall loading of 0.098 MW m⁻². Its neutron budget resolves into a hard 14.03 MeV D–T self-burn line carrying 60 % of the power but 21 % of the count — the shield-setting term — and a soft 2.45 MeV D–D line carrying the rest, both following from a single tritium burnup fraction $f_{\text{burn}} \approx 0.26$. A self-consistent radiation-transport solution fixes the operating point at $T_e = 89.72$ keV. A burn-lean mode cuts the neutron power to the 93.5 MW D–D floor for a 4.1 % fusion-power penalty, a 14.6:1 lever; and the predominantly aneutronic cycle holds a mobile tritium inventory $\sim 275\times$ smaller than a D–T plant,

giving a public dose of $\sim 0.11 \mu\text{Sv h}^{-1}$ at 50 m behind 2.80 m of concrete. A machine that can be shielded to background-scale dose and holds grams, not kilograms, of mobile tritium is a machine that can be sited near people and assets — for both a fixed-defense and an urban housing.

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Reproducibility. The neutron budget, radiation-transport operating point, burn-lean trade, shield curve and tritium source term are frozen anchors (BV-07, KX-L1-A3-BU, AG-CX-16, MV-CX-28, MV-CX-29) in the Kronos Physics De-Risking Register and are regenerable from the neutron-split, triton-kinetics, point-kernel and inventory scripts archived with this paper. This paper is archived at its DOI doi:10.5281/zenodo.22645917 and its official page <https://www.kronosfusionenergy.com/publications/the-neutron-lean-siting-case-for-a-d-he-tandem-mirror-b.html>; the de-risking register is archived at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645917. The neutron-budget arrays, the radiation-transport outputs, the shield curve and the tritium-inventory calculation are archived with the deposit and reference the Kronos 2026 series. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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